

USER MANUAL

WTGAHRS3

GPS IMU





Tutorial Link

[Google Drive](#)

Link to instructions DEMO:

[WITMOTION Youtube Channel](#)

If you have technical problems or cannot find the information that you need in the provided documents, please contact our support team. Our engineering team is committed to providing the required support necessary to ensure that you are successful with the operation of our AHRS sensors.

Contact

[Technical Support Contact Info](#)

Application

- AGV Truck
- Platform Stability
- Auto Safety System
- Movement Measurement
- GPS Positioning
- Car Navigation
- UAV
- Orientation Tracking System
- Displacement Measurement

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1 Introduction

The WTGAHRS3 is a multi-sensor device detecting acceleration, angular velocity, angle, magnetic field as well as GPS. The robust housing and the small outline makes it perfectly suitable for industrial retrofit applications such as condition monitoring and predictive maintenance. Configuring the device enables the customer to address a broad variety of use cases by interpreting the sensor data by smart algorithms.

WTGAHRS3's scientific name is GPS IMU sensor. A sensor measures 3-axis angle, angular velocity, acceleration, magnetic field and GPS data. Its strength lies in the algorithm which can calculate three-axis angle accurately.

WTGAHRS3 is employed where the highest measurement accuracy is required. It offers several advantages over competing sensor:

- Heated for best data availability: new WITMOTION patented zero-bias automatic detection calibration algorithm outperforms traditional accelerometer sensor
- High precision Roll Pitch Yaw (X Y Z axis) Acceleration + Angular Velocity + Angle + Magnetic Field output+GPS data
- Low cost of ownership: remote diagnostics and lifetime technical support by WITMOTION service team
- Developed tutorial: providing manual, datasheet, Demo video, free software for Windows computer, APP for Android smartphones , and sample code for MCU integration including Python, STM32, Arduino, Raspberry Pi, C++, communication protocol for project development
- WITMOTION sensors have been praised by thousands of engineers as a recommended attitude measurement solution

2 Warning Statement

- Putting more than 5 Volt across the sensor wiring of the main power supply can lead to permanent damage to the sensor.
- GPS positioning requires to be operated outdoors.
- For proper instrument grounding: use WITMOTION with its original factory-made cable or accessories.
- Do not access the I2C interface.
- For secondary developing project or integration: use WITMOTION with its compiled sample code.

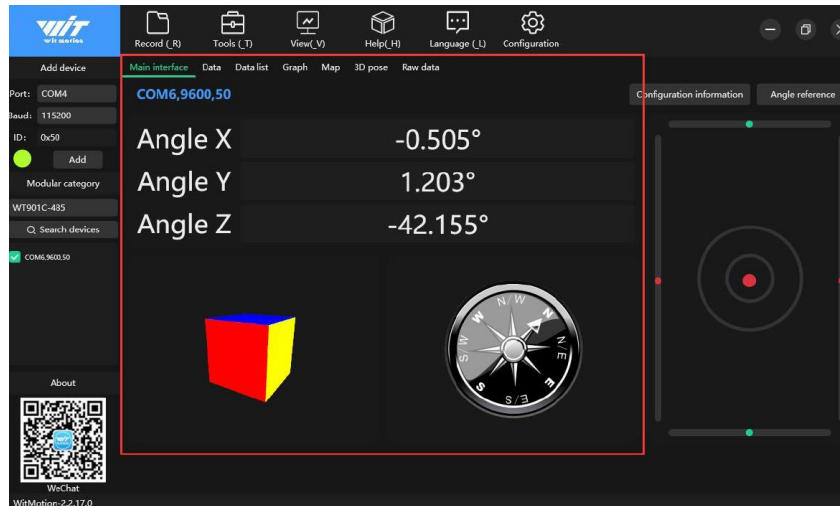
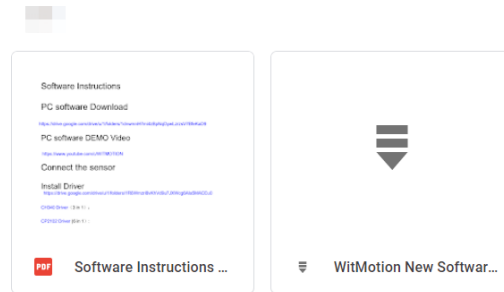
3 Instructions of 2023 New Software

In order to improve the user experience and our customer service, we develop a new version PC software.

Below is the new software and universal instruction download link.

https://drive.google.com/drive/folders/1dnwmnH7mi4zBpNqDywLzrzsV7BfeKaD9?usp=share_link

WITMOTION New Software(Universal)



Note:

The 2022 old version software will be reserved for use. You can check “Chapter 3 Use Instructions with PC” for more details. ↓ ↓ ↓

4 Hardware Connection

4.1 Software Connection

To connect with a computer, a serial module with USB to TTL level is required. We recommend the following two USB to serial modules:



3-in-1 serial converter



6-in-1 serial converter

Step 1. Connect the sensor with a serial converter

PIN Connection:

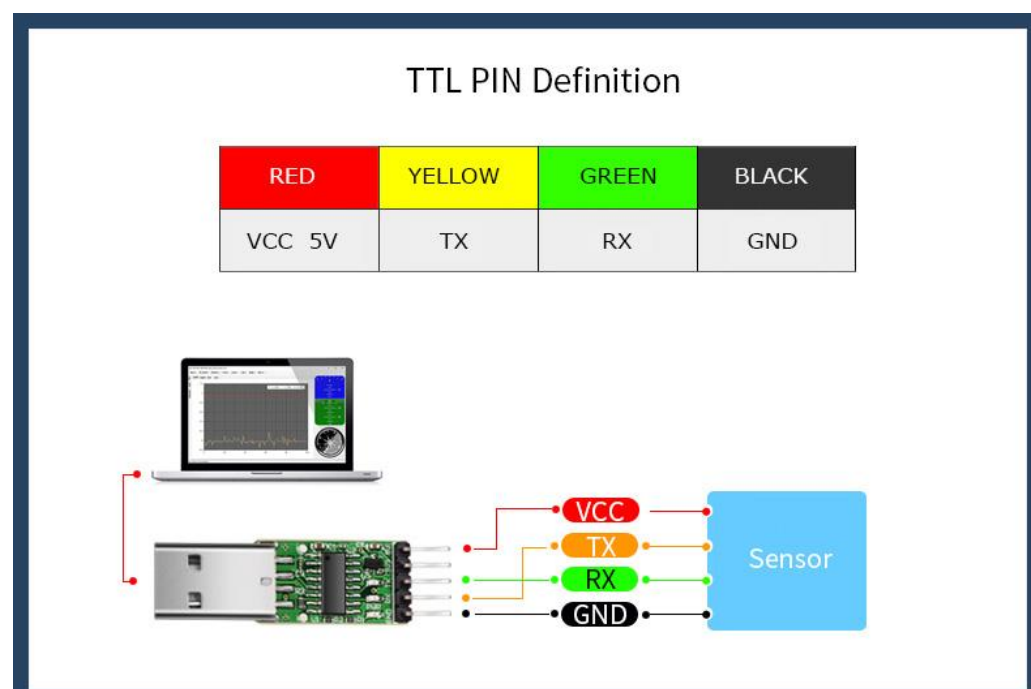
VCC - 5V

TX - RX

RX - TX

GND - GND

(When connecting with computer, VCC-5V is recommended.)



WTGAHRS3| manual v23-0214 | www.wit-motion.com

4.2 Serial Connection

Step 1. Connect the sensor with offered Type-C wire.

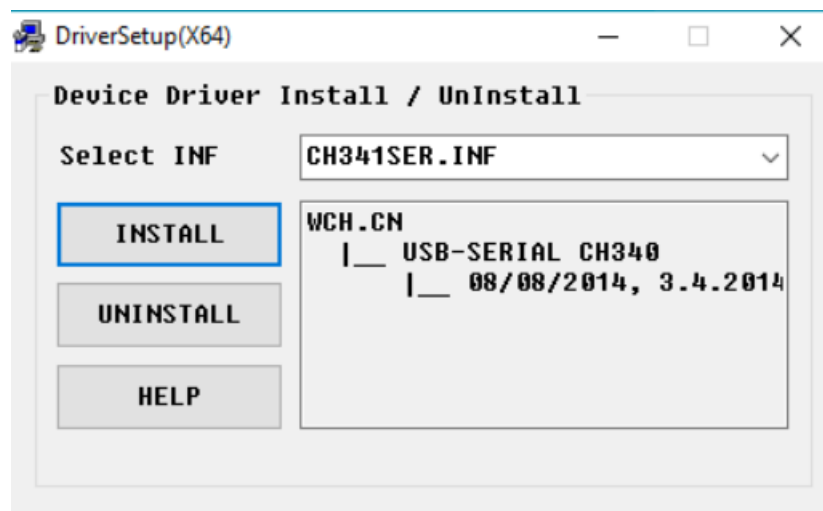
(Warm Reminder: If you wanna use a longer cable, it should be a standard Type-C data cable)

Step 2. Unzip the software and install the driver CH340

<https://drive.google.com/file/d/1I3hl9Thsj9aXfG6U-cQLpV9hC3bVEH2V/view?usp=sharing>

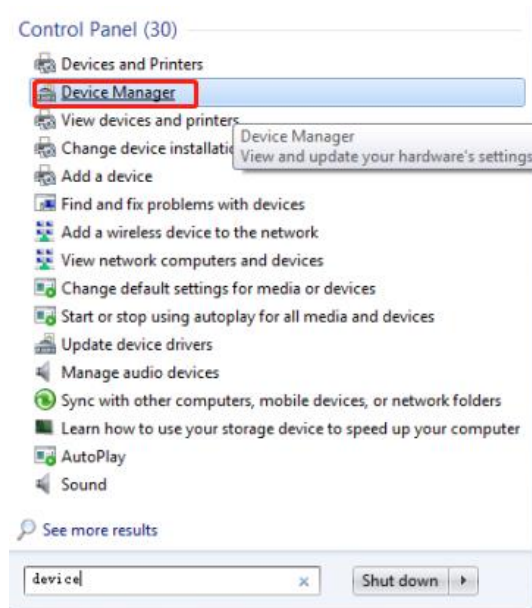
*How to Install and update the CH340 driver

Click the "Uninstall" button first. Then click on the "Install" button.



*How to verify your driver is working

1) To check that the CH340 enumerates to a COM port, you can open the device manager. You can click the **Start** or  (Windows) button and type "*device manager*" to quickly search for the application.



2) After opening the device manager, you will need to open the **Ports (COM & LPT)** tree. The CH340 should show up as **USB-SERIAL CH340 (COM##)**. Depending on your computer, the COM port may show up as a different number.

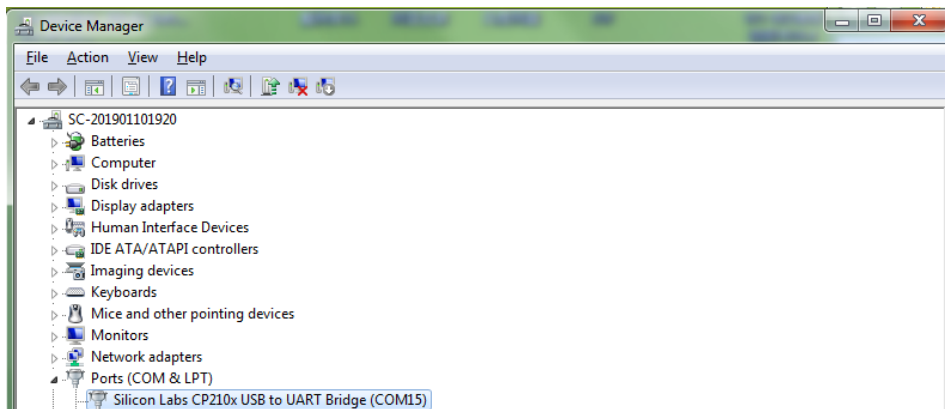




5 Software Introduction

5.1 Instruction

Step 1. Unzip the software and install the driver CH340 or CP2102 (Depending on which accessory for usage.) Installing the CP210X shows the following:



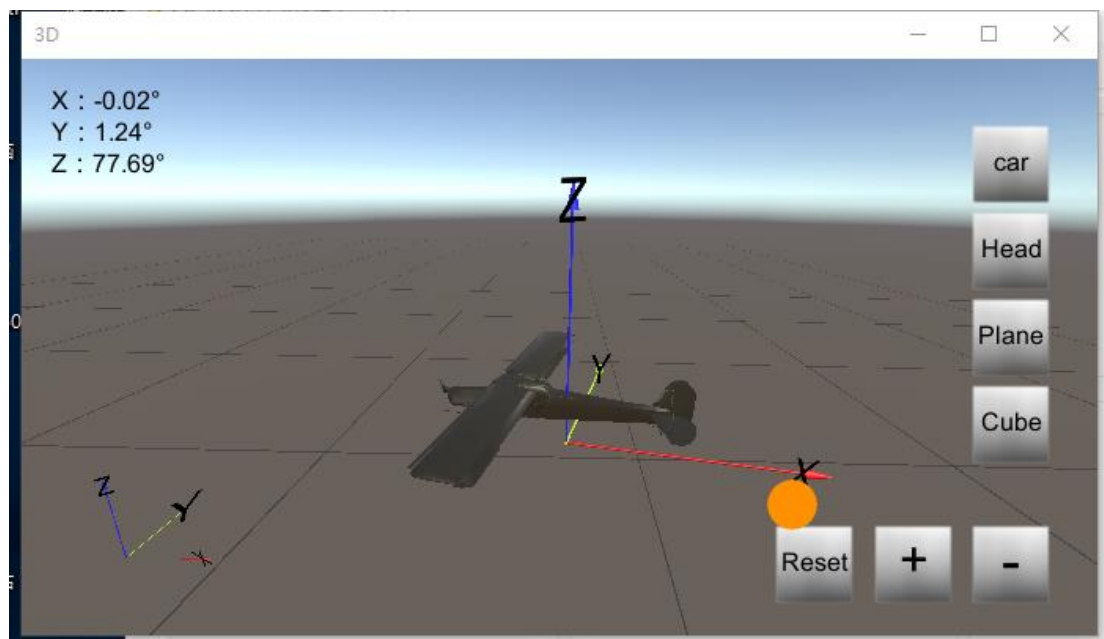
Step 2. Insert the converter to computer and confirm the "com port" in device manager

Step 3. Open the software(Minimu.exe) Data will appear after auto-search finishes.

Step 4. Select “com port”, baud rate 9600 (default)



When the time interval for collecting data is long, the graph is updated slowly. Right-click on the image and click the clear image option to speed up the data refresh rate. Click the 3D button to display the 3D pose of the module.



5.1.1 Main Menu

(Old version software instruction)



Main Menu of software		
Button		Function
File		Launch recorded HEX file (Bin format)
Tools		Hide or display tools box on left side
Record		Record function
3D		3D DEMO
Config		Configuration setting
Help	Language	English or Chinese
	Bluetooth Set	Binding device or unbind
	Firmware update	Option for firmware update
	About Minimu	Info about Minimu.exe
	Factory test	For manufacturer internal test only
Auto-search		Auto searching the sensor
Port		Com port selection
Baud		Baud rate selection
Type		Fixed setting as Normal for WTGAHRS1
Open		Open com port
Close		Close com port

5.1.2 Menu of Configuration

Normal - Config X

Read Config Lock Unlock Calibrate Time

System

Reset Sleep Alarm

Algorithm: 9 - axis
Install Direction: Horizontal
☐ Instruction Startup

Calibrate

Acceleration Magnetic Filed Reset Z-axis Angle

Reset Height Angle Reference

☒ Gyro Auto Calibrate

Range

Acceleration: 16 g
Gyro: 2000 deg/s
Band Width: 20 Hz
GPS Time Zone: UTC-12

Communication

Baud Rate: 9600
Output Rate: 10Hz
Device Address: 0x50 change

Content

☒ Time ☒ Acceleration ☒ Velocity ☒ Angle ☒ Magnetism ☐ Port

☐ Pressure ☐ Location ☐ PDOP ☐ Quaternion ☐ Positioning Accuracy ☐ GPS Original

Port

D0 model: AIN pulse width: 0 ▲ ▼ 0 ▲ ▼ cycle: 0 ▲ ▼

D1 model: AIN pulse width: 0 ▲ ▼ 0 ▲ ▼ cycle: 0 ▲ ▼

D2 model: AIN pulse width: 0 ▲ ▼ 0 ▲ ▼ cycle: 0 ▲ ▼

D3 model: AIN pulse width: 0 ▲ ▼ 0 ▲ ▼ cycle: 0 ▲ ▼

■ Online

Save Config

Read Configuration Completed

Menu of Configuration	
Button	Function
Read Config	Reading the current configuration
Lock	Lock the sensor
Unlock	Unlock the sensor
Calibrate Time	Calibration time of chip
Save Config	Save configuration

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System

Reset Sleep Alarm

Algorithm: 9 - axis

Install Direction: Horizontal

☐ Instruction Startup

Menu of System	
Button	Function
Reset	Reset to factory setting
Sleep	Sleep function
Alarm	Alarm function
Algorithm	6-axis algorithm or 9-axis
Installation Direction	Vertical or horizontal installation
Instruction Start-up	Instructions sending to start-up the sensor

Calibrate

Acceleration Magnetic Field Reset Z-axis Angle ☒ Gyro Auto Calibrate

Reset Height Angle Reference

Menu of Calibrate	
Button	Function
Acceleration	Accelerometer calibration
Magnetic Field	Magnetometer calibration
Reset Height	Reset height data to 0 (only for sensor built-in barometer, including WT901B, WTGAHRS2, HWT901B, WTGAHRS1)
Reset Z-axis Angle	Reset Z-axis angle to 0 degree, not available for WTGAHRS1 in 9-axis algorithm
Angle Reference	Setting current angle as 0 degree
Gyro Auto Calibrate	Auto-calibration of gyroscope

Range

Acceleration: 16 g Gyro: 2000 deg/s Band Width: 20 Hz GPS Time Zone: UTC-12

Menu of Range	
Button	Function
Acceleration	Acceleration measurement range
Gyro	Gyroscope measurement range
Band Width	Bandwidth range
GPS Time Zone	GPS positioning of time zone

Communication

Baud Rate: 9600 Output Rate: 10Hz Device Address: 0x50

Menu of Communication	
Button	Function
Baud Rate	Baud rate selection
Output Rate	Return rate selection
Device Address	Interface for R&D

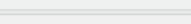
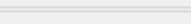
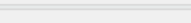
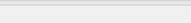
Content

☒ Time
 ☒ Acceleration
 ☒ Velocity
 ☒ Angle
 ☒ Magnetism
 ☐ Port

☐ Pressure
 ☐ Location
 ☐ PDOP
 ☐ Quaternion
 ☐ Positioning Accuracy
 ☐ GPS Original

Menu of Content	
Button	Function
Time	Time data output
Acceleration	Acceleration data output
Velocity	Angular velocity data output
Angle	Angle data output
Magnetism	Magnetic field data output
Port	Port data output
Pressure	Pressure output, only for GPS IMU series, such as WTGAHRS1, WTGAHRS2
Location	Latitude&Longitude data output, only for GPS IMU series, such as WTGAHRS1, WTGAHRS2
PDOP	Ground velocity data output, only for GPS IMU series, such as WTGAHRS1, WTGAHRS2
Quaternion	Quaternion data output
Positioning Accuracy	Option for GPS positioning accuracy output, including Satellite quantity, PDOP, HDOP, VDOP data, only for GPS IMU series, such as WTGAHRS1, WTGAHRS2
GPS Original	Only output GPS raw data, only for GPS IMU series, such as WTGAHRS1, WTGAHRS2
Menu of Port	
D0 Model	Extended port D0
D1 Model	Extended port D1
D2 Model	Extended port D2
D3 Model	Extended port D3
Pulse width	Pulse width of PWM
Cycle	Cycle of PWM

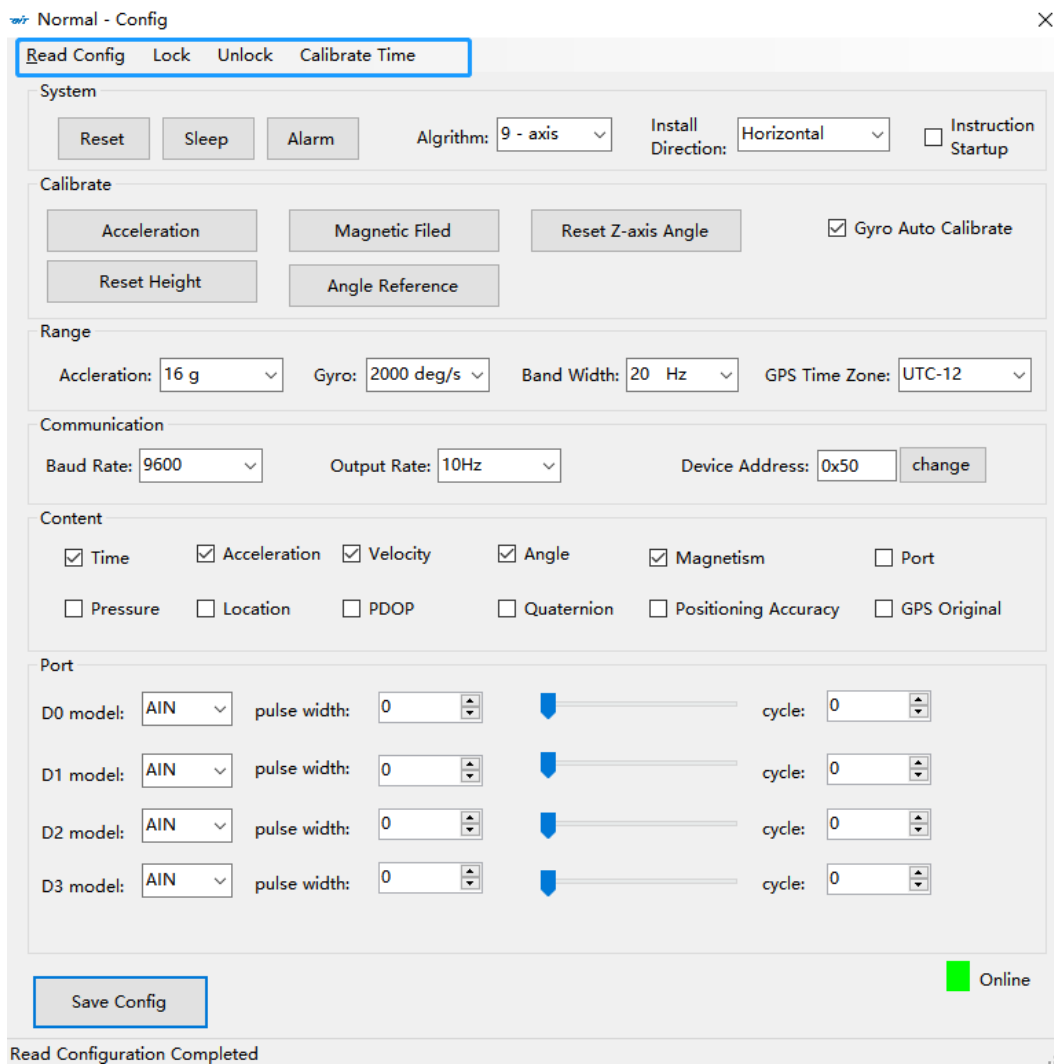
Port

D0 model:	AIN	pulse width:	0			cycle:	0
D1 model:	AIN	pulse width:	0			cycle:	0
D2 model:	AIN	pulse width:	0			cycle:	0
D3 model:	AIN	pulse width:	0			cycle:	0

5.2 Calibration

Notice: The calibration and read configuration should be performed under the online status displayed in the lower right corner of the upper computer configuration column.

As shown in the figure below, offline means that the host computer does not control the module. Before the module can be used, calibration is required. Calibration of the module includes accelerometer calibration.



The screenshot shows the 'Normal - Config' window with the following sections:

- System:** Includes buttons for 'Reset', 'Sleep', and 'Alarm'. It also has 'Algorithm' set to '9 - axis', 'Install Direction' set to 'Horizontal', and an 'Instruction Startup' checkbox.
- Calibrate:** Includes buttons for 'Acceleration', 'Magnetic Filed', 'Reset Z-axis Angle', 'Reset Height', and 'Angle Reference'. There is a checked checkbox for 'Gyro Auto Calibrate'.
- Range:** Includes dropdowns for 'Acceleration' (16 g), 'Gyro' (2000 deg/s), 'Band Width' (20 Hz), and 'GPS Time Zone' (UTC-12).
- Communication:** Includes dropdowns for 'Baud Rate' (9600) and 'Output Rate' (10Hz), and a 'Device Address' field (0x50) with a 'change' button.
- Content:** Includes checkboxes for 'Time', 'Acceleration', 'Velocity', 'Angle', 'Magnetism', 'Port', 'Pressure', 'Location', 'PDOP', 'Quaternion', 'Positioning Accuracy', and 'GPS Original'.
- Port:** Includes four rows for 'D0 model', 'D1 model', 'D2 model', and 'D3 model'. Each row has a dropdown for the model (all set to 'AIN'), a 'pulse width' field (all set to 0), a slider, and a 'cycle' field (all set to 0).
- Buttons:** A 'Save Config' button is at the bottom left, and an 'Online' status indicator (green square) is at the bottom right.
- Status Bar:** At the very bottom, it says 'Read Configuration Completed'.

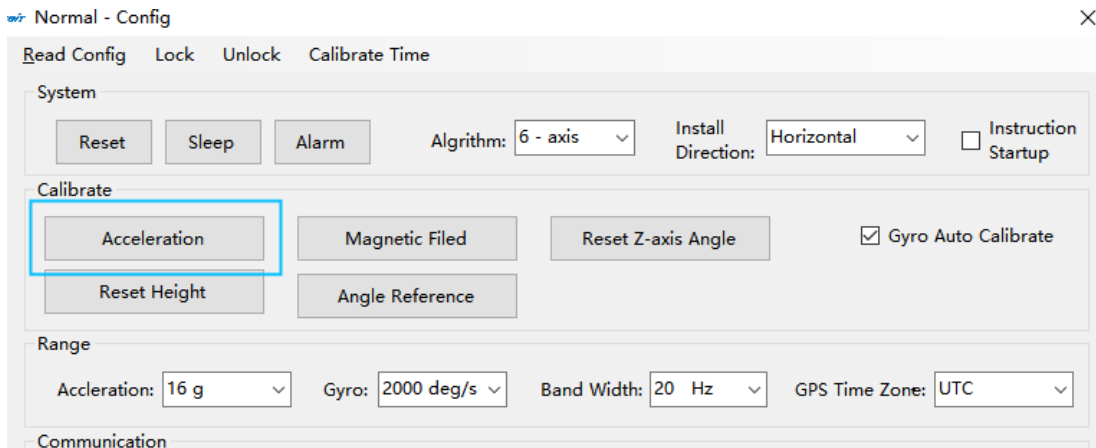
5.2.1 Accelerometer Calibration

Purpose:

The accelerometer calibration is used to remove the zero bias of the accelerometer. Before calibration, there will be different degrees of bias error. After calibration, the measurement will be accurate.

Methods:

- Step 1. Keep the module horizontally stationary
- Step 2. Click the accelerometer calibration
- Step 3. Click the "Start calibration" and wait for 3 seconds



The screenshot shows the 'Normal - Config' window with the 'Calibrate' section highlighted. The 'Calibrate' section contains the following options:

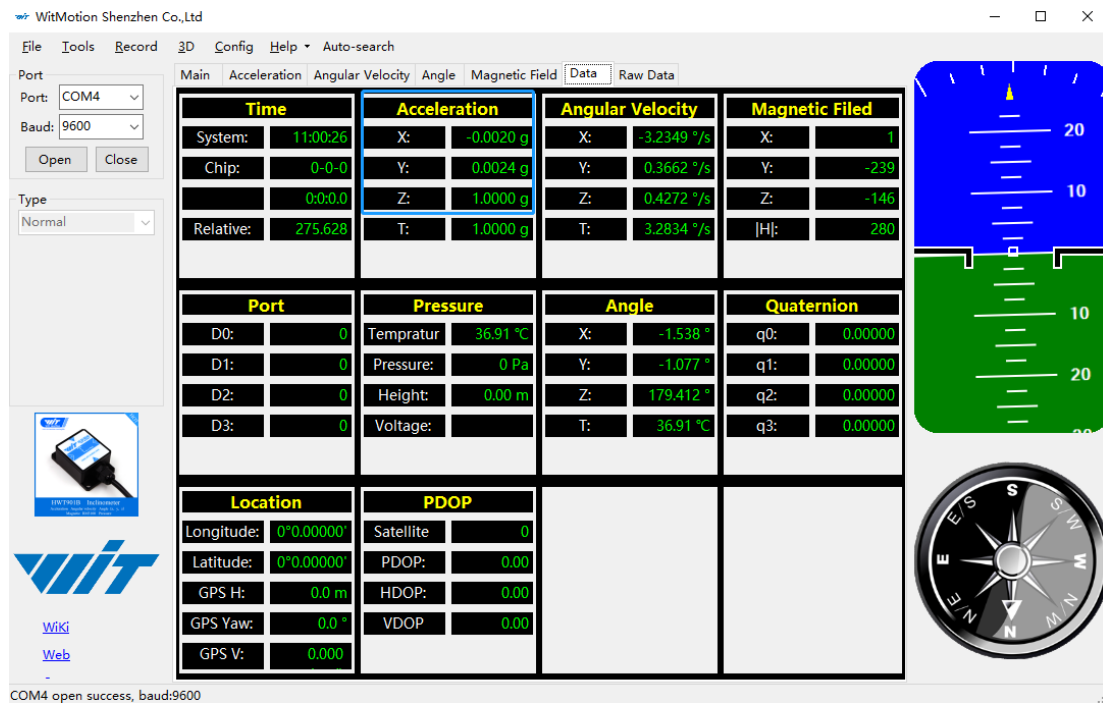
- Acceleration** (highlighted with a blue box)
- Magnetic Filed
- Reset Z-axis Angle
- Reset Height
- Angle Reference
- ☒ Gyro Auto Calibrate

The 'Range' section shows the following settings:

- Acceleration: 16 g
- Gyro: 2000 deg/s
- Band Width: 20 Hz
- GPS Time Zone: UTC

Step 4. Click "Complete Calibration"

Step 5. Judge the result--confirm if there is 1g on Z-axis acceleration



Step 6. After 1 ~ 2 seconds, the three axial acceleration value of the module is about 0, 0, 1, the X and Y axis angle is around 0°. After calibration, the x-y axis Angle is accurate.

Notice: When putting the module horizontal, there is 1g of gravitational acceleration on the Z-axis.

5.2.2 Gyroscope Automatic Calibration

The gyroscope calibration is to calibrate the angular velocity, and the sensor will calibrate automatically.

It is recommended that the automatic calibration of gyroscopes can be inactivated only if the module rotates at a constant speed.

5.3 Configuration

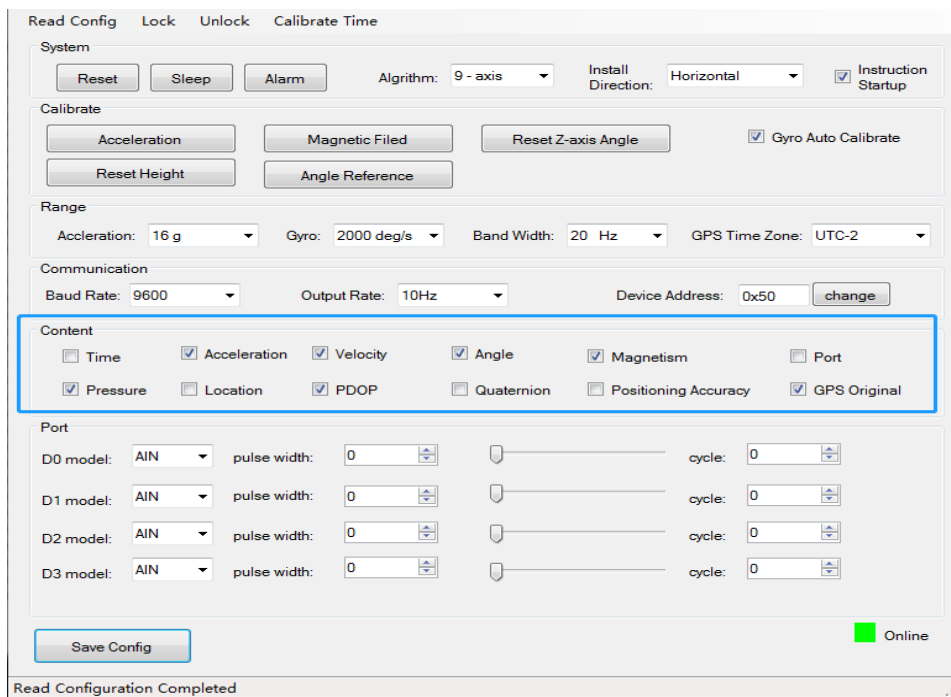
5.3.1 GPS Original Data

Notice:

To set GPS positioning, you need to check whether the GPS has original data output.

Step 1:

Open the configuration bar and click the "GPS original" as output data.



The screenshot shows the WIT motion configuration software interface. The 'Content' section is highlighted with a blue border, showing the following options:

- ☐ Time
- ☒ Acceleration
- ☒ Velocity
- ☒ Angle
- ☒ Magnetism
- ☐ Port
- ☒ Pressure
- ☐ Location
- ☒ PDOP
- ☐ Quaternion
- ☐ Positioning Accuracy
- ☒ GPS Original

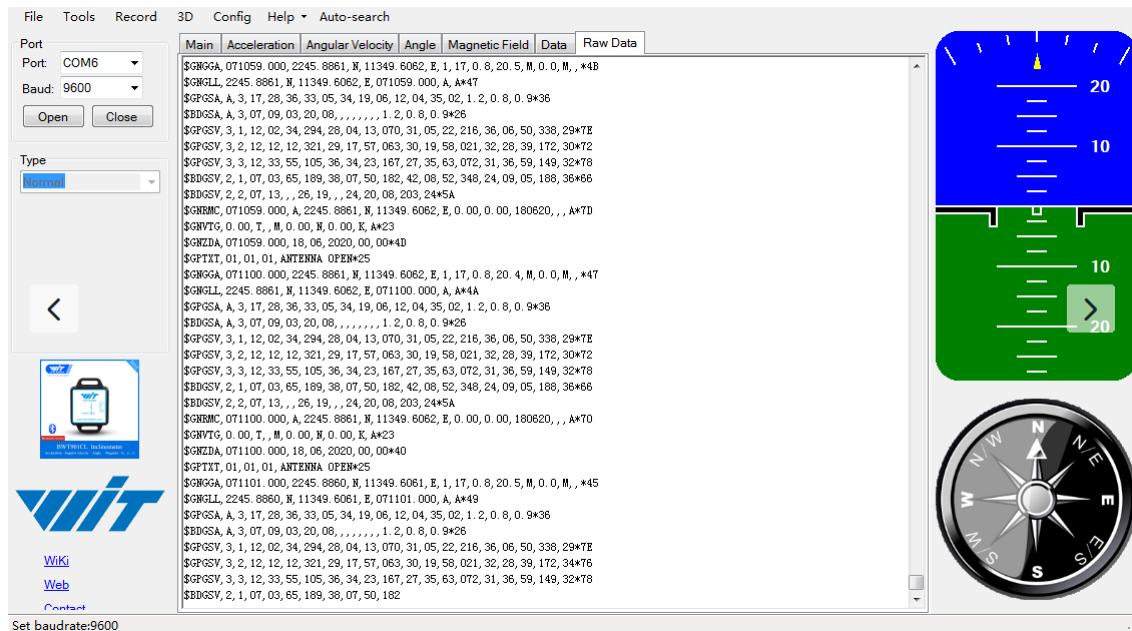
Other visible settings include:

- System:** Reset, Sleep, Alarm, Algorithm: 9-axis, Install Direction: Horizontal, Instruction Startup: ☒
- Calibrate:** Acceleration, Magnetic Filed, Reset Z-axis Angle, Gyro Auto Calibrate: ☒, Reset Height, Angle Reference
- Range:** Acceleration: 16 g, Gyro: 2000 deg/s, Band Width: 20 Hz, GPS Time Zone: UTC-2
- Communication:** Baud Rate: 9600, Output Rate: 10Hz, Device Address: 0x50 (change)
- Port:** D0 model: AIN, pulse width: 0, cycle: 0; D1 model: AIN, pulse width: 0, cycle: 0; D2 model: AIN, pulse width: 0, cycle: 0; D3 model: AIN, pulse width: 0, cycle: 0
- Buttons:** Save Config, Online (green indicator)
- Status Bar:** Read Configuration Completed

Step 2:

The PC software will display the original GPS output data.

Confirm if there is NMEA-0183 GPS raw data ensuring positioning is actually working.



Notice:

If the device showing as offline, try to click other data as output data, and then reopen the configuration to see if it is online.

Step 2:

After the verification is completed, unclick "GPS original".

Otherwise, the module only outputs the original GPS data and no other data.

5.3.2 Return Content

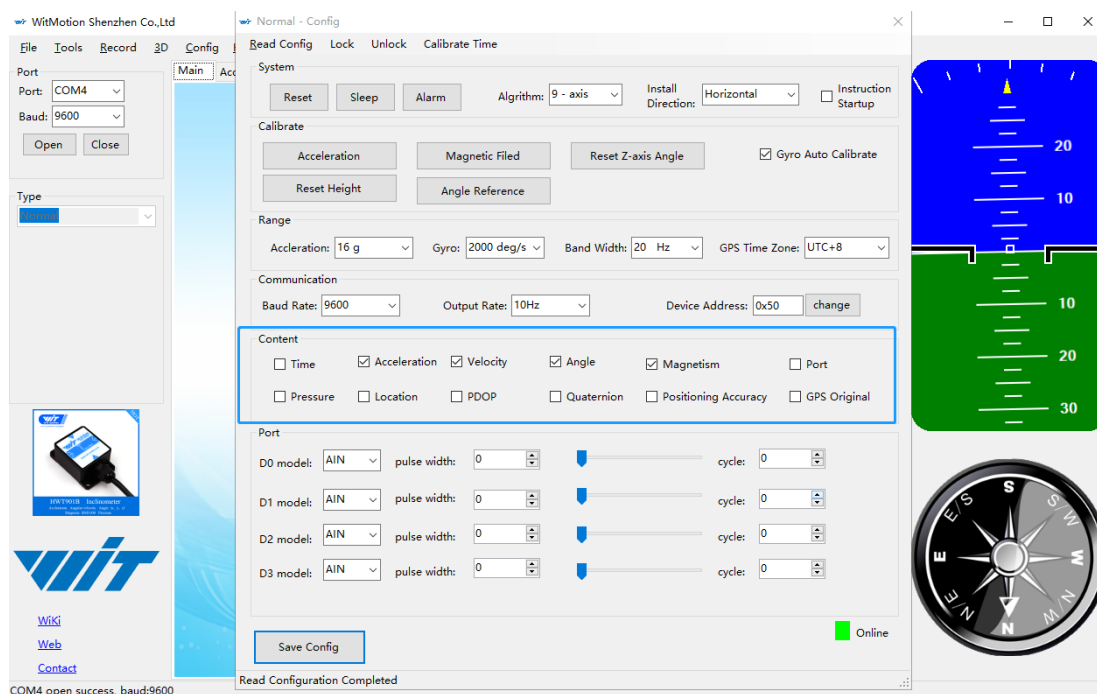
Setting method: The content of the data return can be set according to user needs, click the configuration option bar, and check the data content to be output.

WTGAHRS3, the default output is acceleration, angular velocity, angle. The time is the internal time of the module. The default is that the initial time of power-on is 0:0:0.0 on January 1, 2015. If the GPS module is connected, the time is the GPS receiving time.

GPS time will be 8 hours later than Beijing time.

To get the correct data, you need to check the "Location", "PDOP", and "Positioning Accuracy" in the settings.

Notice: After selecting "GPS original", the module will only output the original GPS information, and no other data will be output.



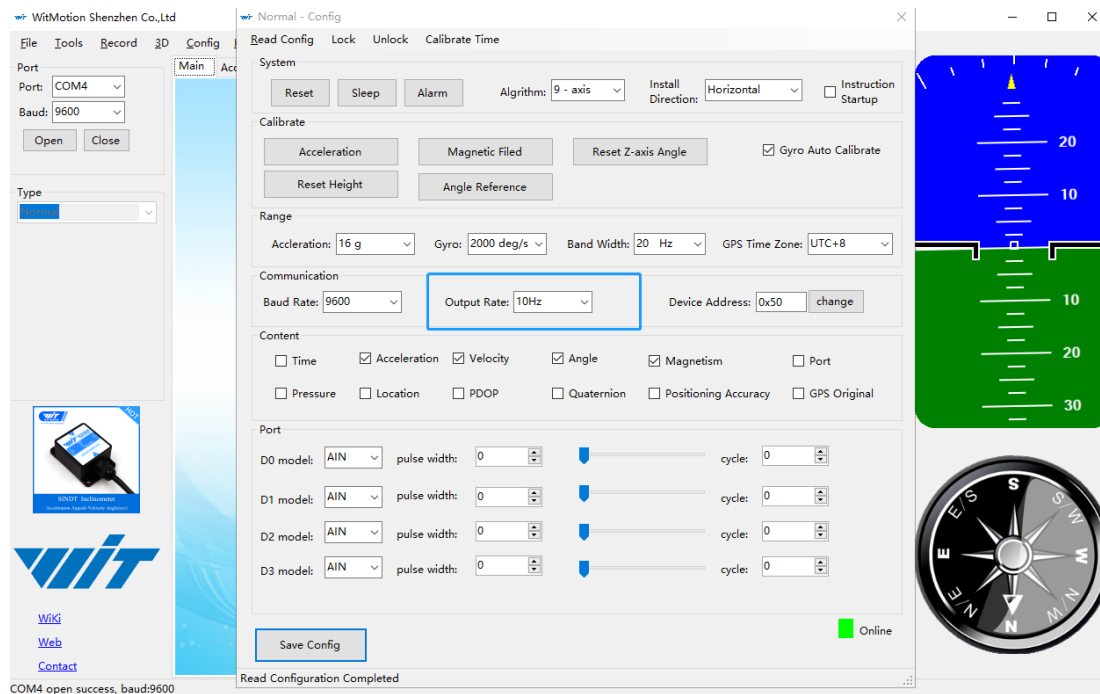
After a few minutes of successful GPS positioning, the GPS information will be displayed.

5.3.3 Output Rate

Method: Click the configuration option bar, and select the return rate from 0.1 to 200HZ in the configuration column.

The default return rate of the module is 10Hz, the highest return rate supports 200Hz. The return rate of the GPS module is 1 Hz, which cannot be changed.

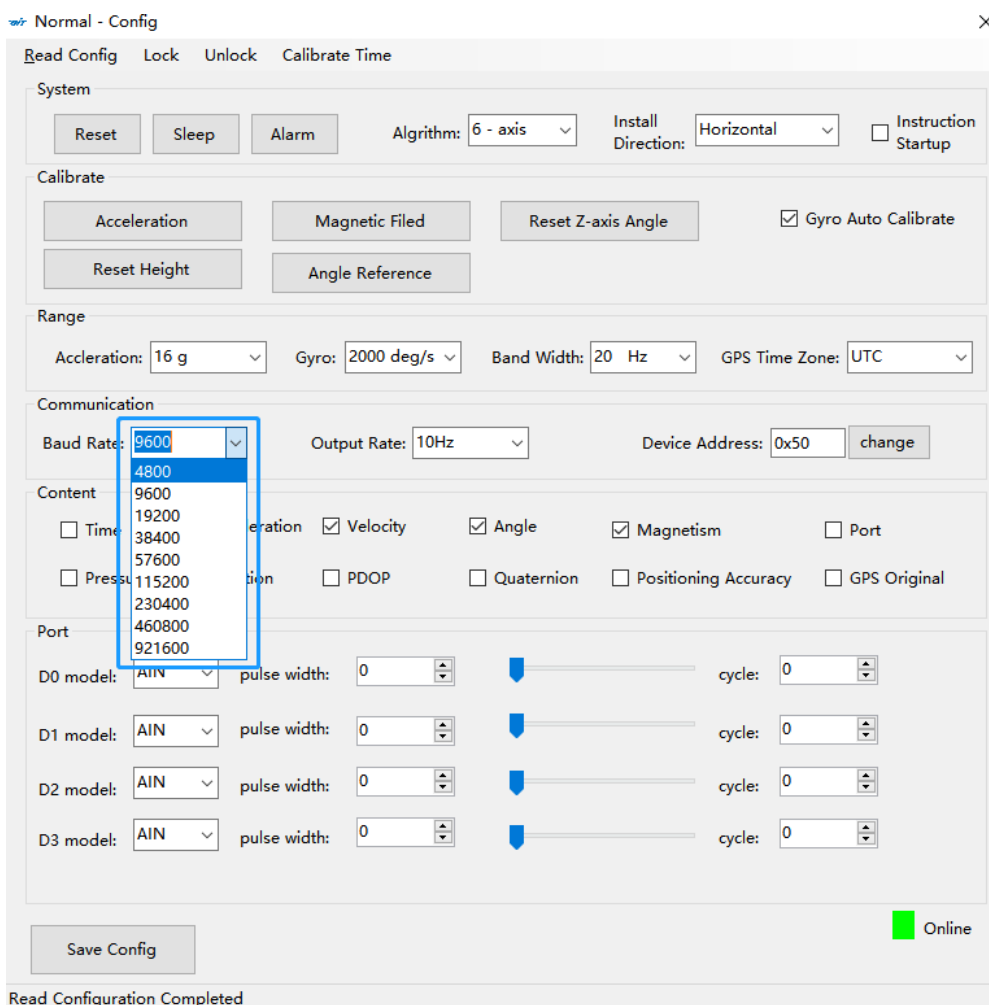
Notice: If there are more backhaul contents and the communication baud rate is lower, it may not be possible to transmit so much data. Then the module will automatically reduce the frequency and output at the maximum allowable output rate. To put it simply, if the return rate is high, the baud rate should also be set higher, generally 115200.



5.3.4 Baud Rate

The module supports multiple baud rates, and the default baud rate is 9600. To set the baud rate of the module, you need to select the baud rate to be changed in the communication rate drop-down box in the configuration bar based on the correct connection between the software and the module.

Note: After the change, the module will no longer output data at the original baud rate. The data will be output only when the baud rate that has been changed is selected on the PC software again.

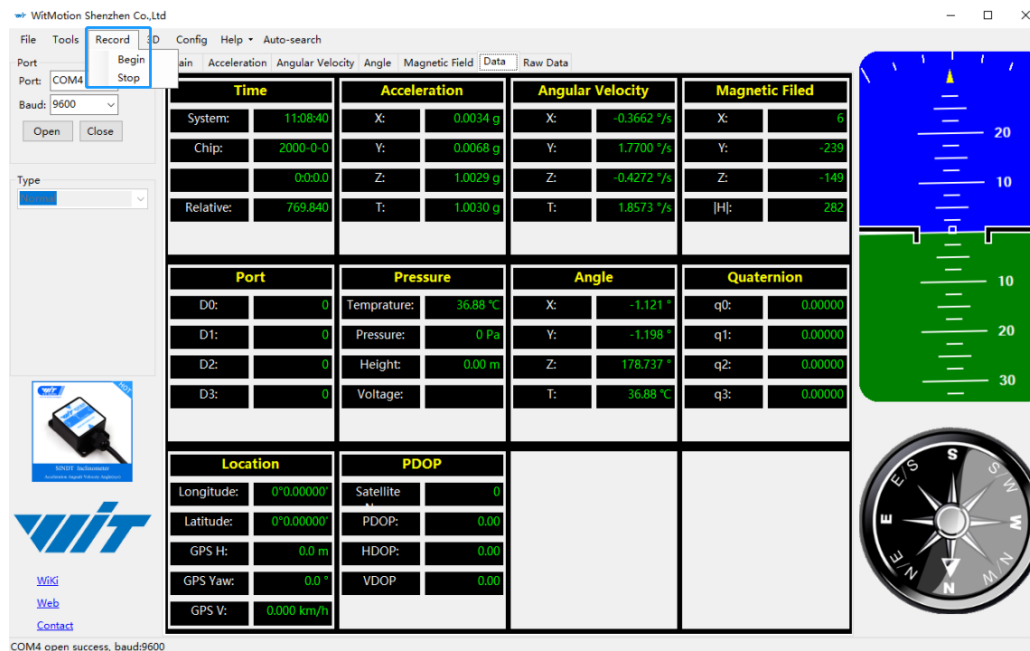


The screenshot shows the 'Normal - Config' window with the 'Communication' section selected. The 'Baud Rate' dropdown menu is open, showing a list of available baud rates: 9600, 4800, 19200, 38400, 57600, 115200, 230400, 460800, and 921600. The '9600' option is currently selected. Other settings visible include 'Output Rate' set to 10Hz, 'Device Address' set to 0x50, and various checkboxes for data output like Velocity, Angle, Magnetism, etc. The status bar at the bottom indicates 'Read Configuration Completed' and 'Online'.

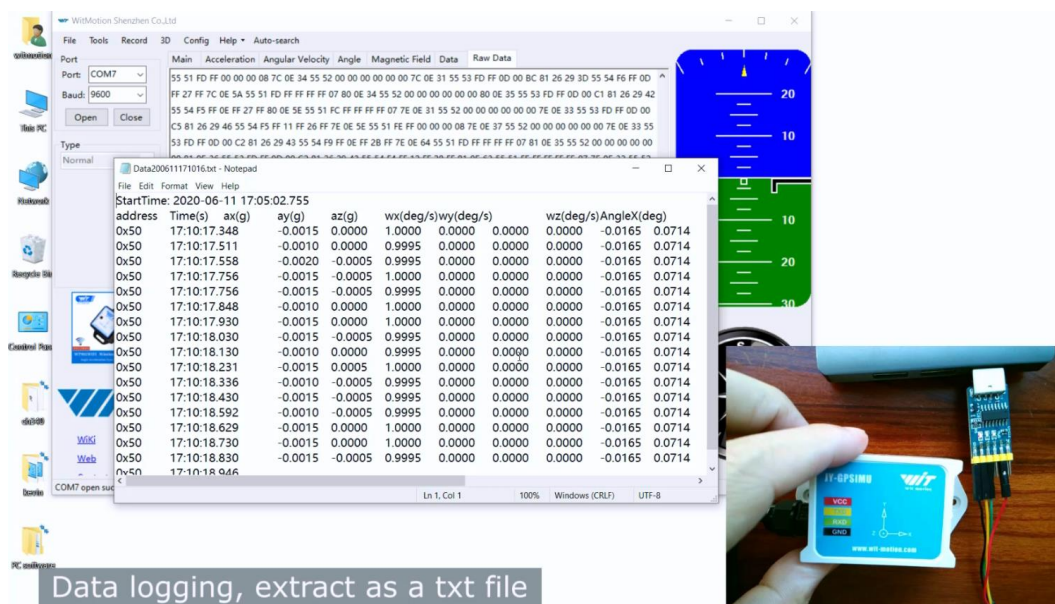
5.3.5 Data Recording

Method are as follows:

Step 1: Click "Record" and "Begin"



Step 2: Click "Stop"



Step 3: Extract the data as "txt" file

```

StartTime: 2021-05-06 15:33:29.138
address  Time(s)  ax(g)  ay(g)  az(g)  wx(deg/s)  wy(deg/s)  wz(deg/s)  AngleX(deg)  AngleY(deg)  AngleZ(deg)  T(°)
Dx50    15:33:29.138  0.2900 -0.0112 0.9985 0.0000 0.0000 0.0000 -0.6921 -16.1609 0.4834 32.42 0 0
Dx50    15:33:29.279  0.2891 -0.0117 0.9980 0.0000 0.0000 0.0000 -0.6812 -16.1719 0.4834 32.42 0 0
Dx50    15:33:29.339  0.2891 -0.0117 0.9980 0.0000 0.0000 0.0000 -0.6812 -16.1609 0.4834 32.41 0 0
Dx50    15:33:29.456  0.2886 -0.0117 0.9980 0.0000 0.0000 0.0000 -0.6812 -16.1554 0.4834 32.39 0 0
Dx50    15:33:29.548  0.2881 -0.0122 0.9980 0.0000 0.0000 0.0000 -0.6866 -16.1499 0.4834 32.44 0 0
Dx50    15:33:29.626  0.2896 -0.0122 0.9990 0.0000 0.0000 0.0000 -0.6866 -16.1499 0.4834 32.38 0 0
Dx50    15:33:29.753  0.2881 -0.0127 0.9980 0.0000 0.0000 0.0000 -0.6866 -16.1609 0.4834 32.44 0 0
Dx50    15:33:29.864  0.2891 -0.0103 1.0005 0.0000 0.0000 0.0000 -0.6976 -16.1609 0.4834 32.41 0 0
Dx50    15:33:29.927  0.2900 -0.0132 0.9976 0.0000 0.0000 0.0000 -0.6702 -16.1609 0.4834 32.44 0 0
Dx50    15:33:30.101  0.2881 -0.0098 0.9980 0.0000 0.0000 0.0000 -0.6921 -16.1719 0.4834 32.39 0 0
Dx50    15:33:30.132  0.2896 -0.0137 0.9956 0.0000 0.0000 0.0000 -0.6866 -16.1609 0.4834 32.41 0 0
Dx50    15:33:30.278  0.2891 -0.0112 0.9980 0.0000 0.0000 0.0000 -0.6812 -16.1554 0.4834 32.42 0 0
Dx50    15:33:30.336  0.2900 -0.0127 0.9966 0.0000 0.0000 0.0000 -0.6921 -16.1609 0.4834 32.42 0 0
Dx50    15:33:30.457  0.2891 -0.0132 0.9980 0.0000 0.0000 0.0000 -0.6757 -16.1554 0.4834 32.44 0 0
Dx50    15:33:30.549  0.2881 -0.0107 0.9980 0.0000 0.0000 0.0000 -0.7031 -16.1664 0.4834 32.44 0 0
Dx50    15:33:30.627  0.2900 -0.0122 0.9980 0.0000 0.0000 0.0000 -0.6866 -16.1554 0.4834 32.42 0 0
Dx50    15:33:30.748  0.2891 -0.0117 0.9976 0.0000 0.0000 0.0000 -0.6921 -16.1664 0.4834 32.44 0 0
Dx50    15:33:30.842  0.2896 -0.0117 0.9966 0.0000 0.0000 0.0000 -0.6812 -16.1719 0.4834 32.41 0 0
Dx50    15:33:30.920  0.2896 -0.0132 0.9966 0.0000 0.0000 0.0000 -0.6812 -16.1499 0.4834 32.38 0 0
Dx50    15:33:31.098  0.2881 -0.0107 0.9966 0.0000 0.0000 0.0000 -0.6976 -16.1609 0.4834 32.44 0 0

```

It is highly recommended that data can be pasted to an Excel file. In this way, all data will be shown in order.

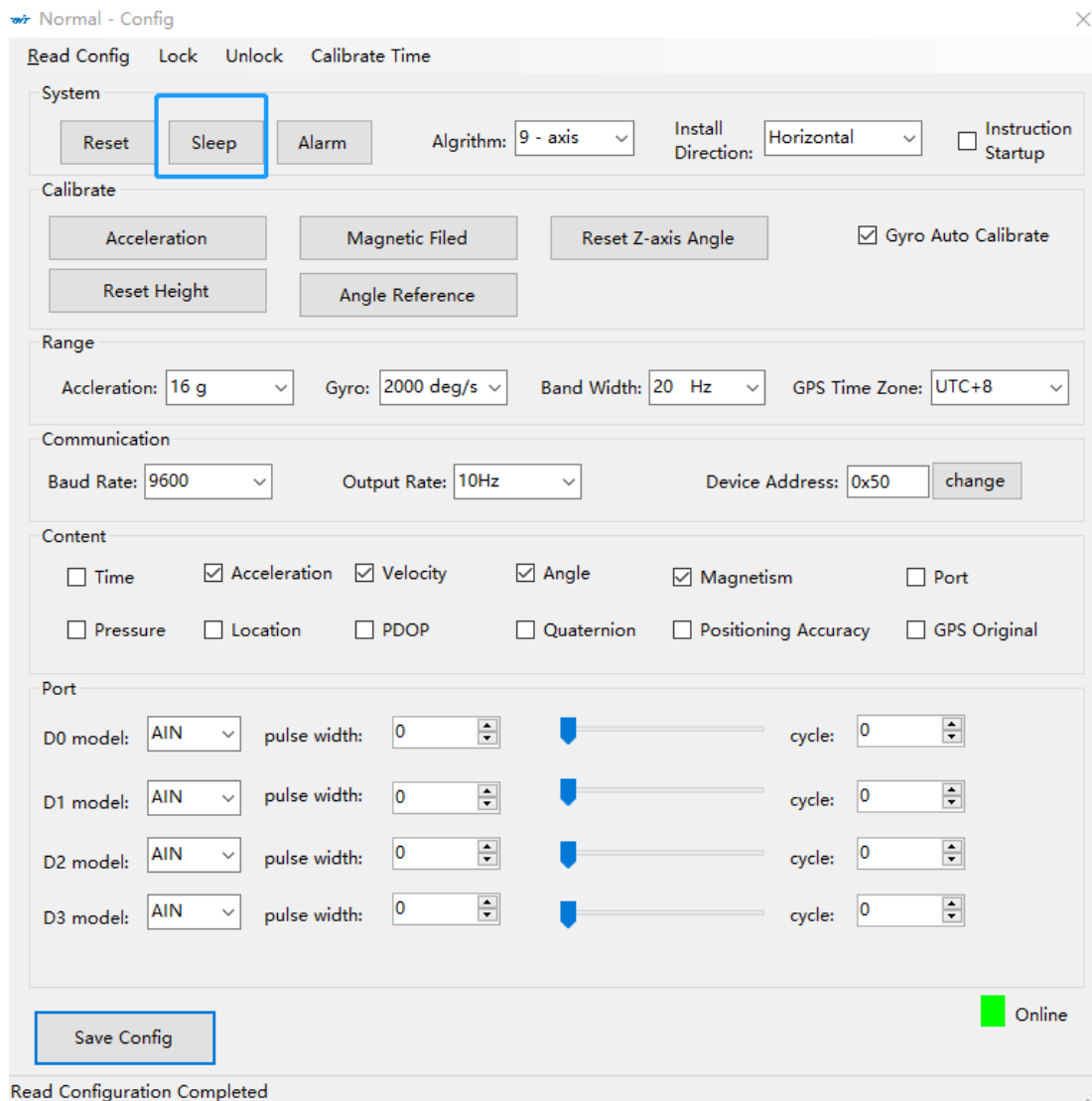
Time represents time, ax ay az represents the acceleration in the three axes of x y z, wx wy wz represents the angular velocity in the three axes of x y z. Anglex Angley Anglez represents the angles of the three axial directions of x y z, T represents the temperature.

5.3.6 Sleep and Wake Up

Sleep: The module paused working and entered the standby mode. Power consumption is reduced after sleeping.

Wake up: The module enters the working state from the standby state.

The module defaults to a working state, in the "Config" of the software, click "Sleep" option to enter the sleep state, click "Sleep" again to release sleep.



Normal - Config

Read Config Lock Unlock Calibrate Time

System

Reset **Sleep** Alarm Algorithm: 9 - axis Install Direction: Horizontal ☐ Instruction Startup

Calibrate

Acceleration Magnetic Filed Reset Z-axis Angle ☒ Gyro Auto Calibrate

Reset Height Angle Reference

Range

Acceleration: 16 g Gyro: 2000 deg/s Band Width: 20 Hz GPS Time Zone: UTC+8

Communication

Baud Rate: 9600 Output Rate: 10Hz Device Address: 0x50

Content

☐ Time ☒ Acceleration ☒ Velocity ☒ Angle ☒ Magnetism ☐ Port

☐ Pressure ☐ Location ☐ PDOP ☐ Quaternion ☐ Positioning Accuracy ☐ GPS Original

Port

D0 model: AIN pulse width: 0 cycle: 0

D1 model: AIN pulse width: 0 cycle: 0

D2 model: AIN pulse width: 0 cycle: 0

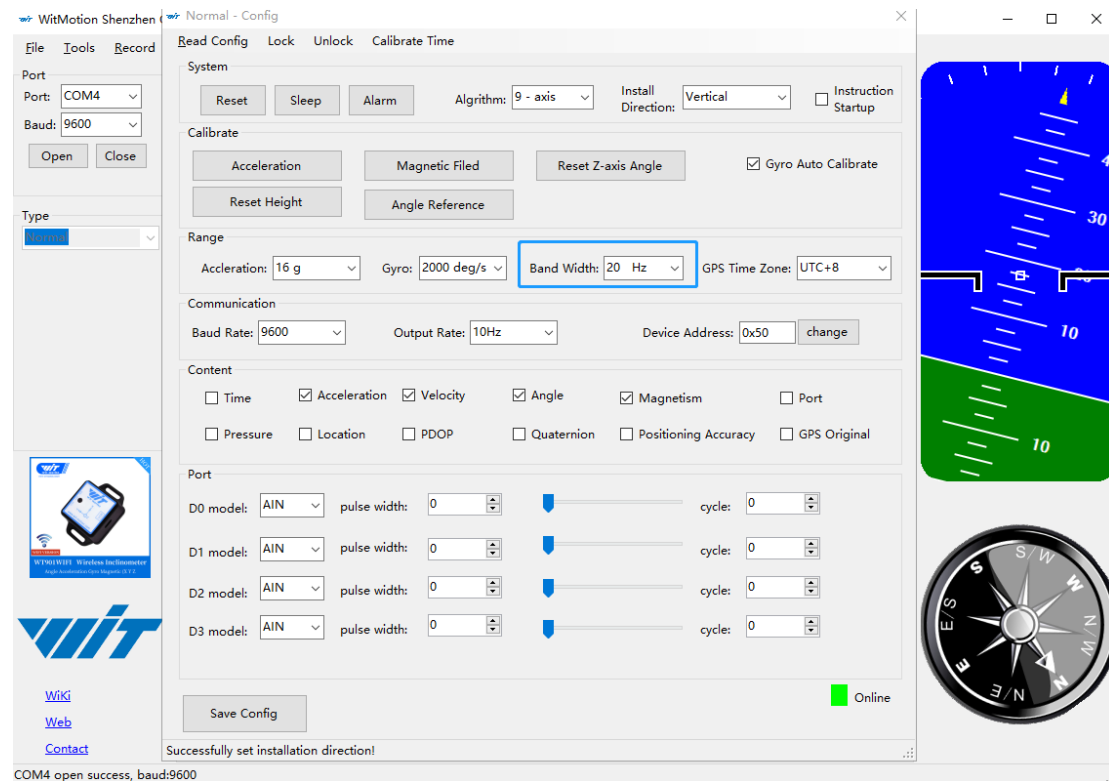
D3 model: AIN pulse width: 0 cycle: 0

Online

Read Configuration Completed

5.3.7 Bandwidth

Default bandwidth is 20Hz.



The module only outputs data within the measurement bandwidth.
Data larger than the bandwidth will be automatically filtered out.

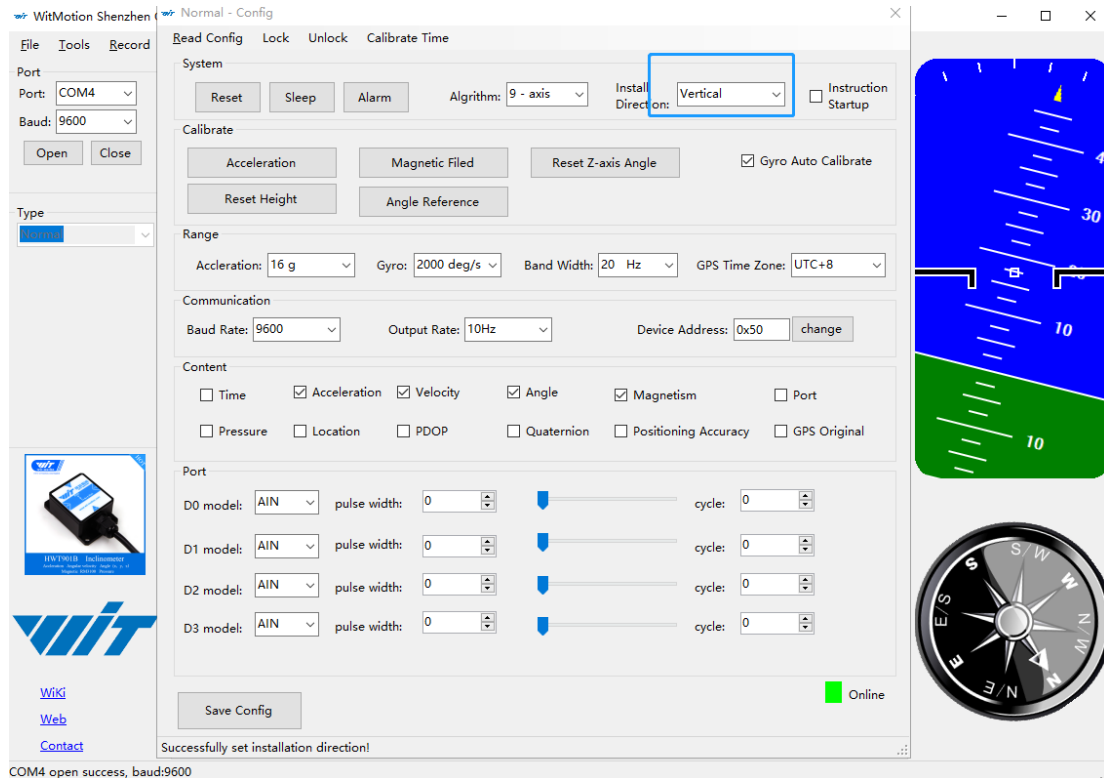
5.3.8 Installation Direction

The default installation direction of the module is horizontal. When the module needs to be installed vertically, the vertical installation can be set.

Step 1: Rotate the module 90 degrees around the X-axis

Step 2: Place the sensor 90 degrees vertically

Step 3: Click "Vertical" as install directions on "Config" menu



6 Sample code

[Link to download all sample code](#)